Data Management Project

Project Report

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Prof

SUBMITTED BY

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FIRST YEAR

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Revision History

S#	Version	Contributors	Date	Description of change
1	Version 1	Fayyaz Hussain Shah	30/11/2025	Document created
2	Version 2	Fayyaz Hussain Shah	1/12/2025	Document Revised

1 Introduction

1.1 Problem Background

Understanding the well-being of nations is becoming increasingly important for policy making, global development, and social research. The World Happiness Report provides annual scores for countries based on income, health, freedom, social support and other factors.

However, the publicly available World Happiness dataset is often limited to specific years (e.g., 2006–2020 in Kaggle). To perform long-term trend analysis, it becomes necessary to collect additional years from an external API, merge them, assess data quality, and store them properly.

This project builds a full ETL (Extract–Transform–Load) pipeline for the World Happiness dataset, integrating CSV and API sources into a unified analysis-ready dataset.

1.2 Project Goals

- Collect historical data from a CSV dataset (2006–2020)
- Collect recent data from an API source (2021–2024)
- Integrate both datasets into a single standardized dataset
- Assess and improve data quality
- Store the final dataset in SQL and NoSQL databases
- Perform Exploratory Data Analysis (EDA)
- Identify trends in Life Ladder and Log GDP per capita

1.3 Research Questions

- What is the relationship between income (Log GDP) and happiness (Life Ladder)?
- Top Performers of Life Ladder and GDP
- Evolution of GDP and Life Ladder Globally
- Evolution of Italy’s Life Ladder and GDP throughout.

2 Data Acquisition

2.1 Source 1: Kaggle Dataset	2.2 Source 2: Open API World Bank
https://www.kaggle.com/datasets/ajaypalsinghlo/world-happiness-report-2021	http://api.worldbank.org/v2/country/all/indicator/NY.GDP.PCAP.CD?date=2020:2025&format=json&per_page=20000

`df = pd.read_csv('world-happiness-report.csv')`	`# Years 2020-2025`
	`start_year = 2020`
	`end_year = 2025`
	
	`# API URL for GDP per capita for all countries`
	`api_url = (f"http://api.worldbank.org/v2/country/all/indicator/NY.GDP"`
	`f".PCAP.CD?date={start_year}:{end_year}&format=json&per_page=20000")`
	
	`response = requests.get(api_url)`
	`data = response.json()`
	
	`# Extract relevant data`
	`records = []`
	`for entry in data[1]:  # data[0] has metadata`
	`country = entry['country']['value']`
	`year = int(entry['date'])`
	`gdp_per_capita = entry['value']  # Can be None`
	`records.append({`
	`'Country name': country,`
	`'year': year,`
	`'GDP per capita': gdp_per_capita`
	`})`
	
	`df_api = pd.DataFrame(records)`

3 Data Quality Assessment & Enrichment -Pre Integration

Source 1 Kaggle Dataset	Source 2 API Data
3.1 Completeness - Handling Missing Values	
`df.isnull().sum()` `df['Log GDP per capita'] = df['Log GDP per capita'].fillna(df['Log GDP per capita'].median())`	`# Drop missing GDP values` `df_api = df_api.dropna(subset=['GDP per capita'])`

`def fill_missing_years(df):`
`  # Find the range of years`
`  df['year'] = df['year'].astype(int)`
 	
`  # Get full range of years in dataset`
`  year_min = df['year'].min()`
`  year_max = df['year'].max()`
 	
`  # Get list of countries`
`  countries = df['Country name'].unique()`
 	
`  # Create full grid`
`  full_grid = pd.MultiIndex.from_product(`
`  [countries, range(year_min, year_max + 1)],`
`  names=['Country name', 'year']`
`  ).to_frame(index=False)`
 	
`  # Left merge ensures all countries x years exist`
`  df_full = full_grid.merge(df, on=['Country name', 'year'], how='left')`
 	
`  # Sort by country and year`
`  df_full = df_full.sort_values(['Country name', 'year'])`
 	
`  # Numeric columns to fill`
`  numeric_cols = ['Life Ladder', 'Log GDP per capita', 'Social support',`
`  'Healthy life expectancy at birth', 'Freedom to make life choices',`
`  'Generosity', 'Perceptions of corruption']`
 	
`  # Interpolate missing values`
`  df_full[numeric_cols] = df_full.groupby('Country name')[numeric_cols].transform(lambda g: g.interpolate())`
 	
`  # Forward/backward fill if first/last years were missing`
`  df_full[numeric_cols] = df_full.groupby('Country name')[numeric_cols].transform(lambda g: g.ffill().bfill())`
`  return df_full`	

3.2 Consistency - Handling Duplication & Formatting	

`df = df.drop_duplicates()`
 	
`df['Country name'] = df['Country name'].str.strip()`
 	
`df['Country name'] = df['Country name'].str.title()`
 	
`df["year"] = df["year"].astype(int)`
`[25]`
 	
`df = df[['Country name', 'year', 'Life Ladder', 'Log GDP per capita']]`
	`# Calculate Log GDP per capita`
	`df_api['Log GDP per capita'] = np.log(df_api['GDP per capita'])`
	
	`# Drop original GDP per capita if not needed`
	`df_api = df_api[['Country name', 'year', 'Log GDP per capita']]`

4 Data Integration

```
# Integration
df_merged = pd.concat([df, df_api], ignore_index=True)

# Sorting
df_merged = df_merged.sort_values(['Country name', 'year']).reset_index(drop=True)
```

5 Data Quality Assessment & Enrichment -Post Integration

```
df_merged.isnull().sum()
df_merged['Life Ladder'] = df_merged.groupby('Country name')['Life Ladder'].transform(lambda x: x.interpolate())
df_merged['Life Ladder'] = df_merged['Life Ladder'].fillna(df['Life Ladder'].median())
```

```
df_merged.isna().mean()*100
[26]
Country name          0.0
year                  0.0
Life Ladder           0.0
Log GDP per capita    0.0
dtype: float64
```

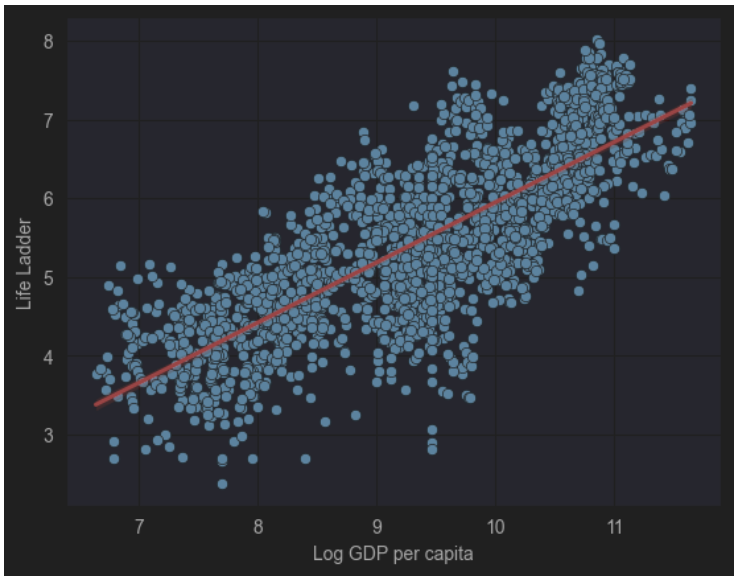
6 Database Storage - SQL

```
conn = sqlite3.connect("WHR.db")
df_merged.to_sql("World_Happiness_Report", conn, if_exists="replace", index=False)
conn.close()
```

7 Exploratory Data Analysis

7.1 Relationship of the GDP of a country and the happiness of its people

```
sns.scatterplot(x='Log GDP per capita', y='Life Ladder', data=df_full)
sns.regplot(x='Log GDP per capita', y='Life Ladder', data=df, scatter=False, color='red')
plt.show()
```



Strength of relationship

The closeness of the points to the regression line shows the strength and the opposite reflects weakness.

Direction of relationship

The upward slope shows partial direct proportionality – partial because of the not so strong relationship

Exceptions – Outliers

Countries above the red line: happier than expected for their GDP

Countries below the line: less happy than expected for their GDP

Summary

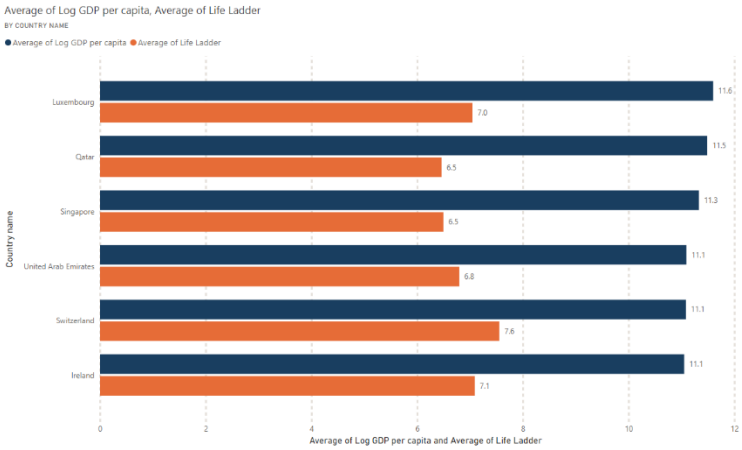
Scatter + regression line shows how Life Ladder changes with Log GDP per capita.

Each point represents a country-year, and the red line shows the average trend.

Points close to the line mean strong relationship; outliers show exceptions.

7.2 Top Performers of GDP and Life Ladder

These visual studies the top performing countries throughout the years and see if they are commonly performing for both or not.



The top performing countries do show a strong positive connection between the GDP and Happiness score however within them there are many contradictory outliers

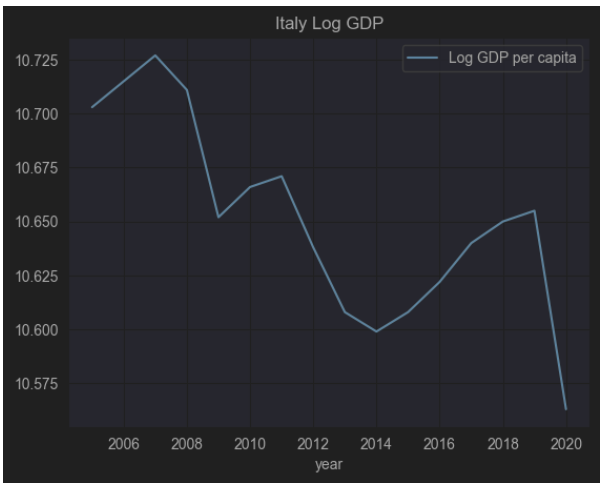
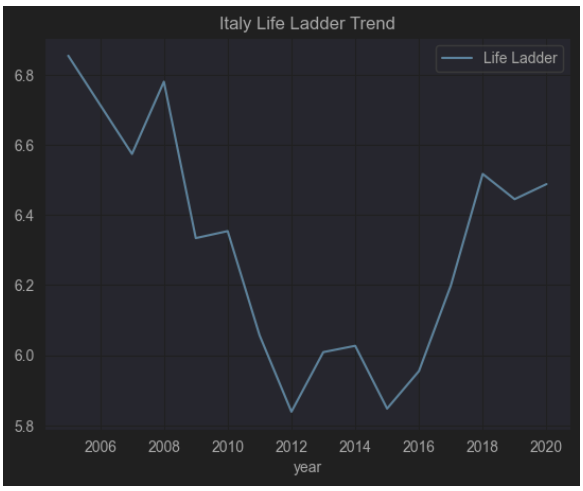
There are various Middle East countries who couldn't achieve the same happiness score even after matching the GDP with the top runners

This concludes that though Happiness score has strong ties with the GDP but it is not the only contributing factor

Visual 2

NOTE: The Graph on top is incomplete. Please click on the link above to view the complete

7.3 Evolution of Life Ladder and GDP in Italy thought out the time

```
country = "Italy"
df[df["Country name"] == country].plot(x="year", y="Life Ladder", title=f"{country} Life Ladder Trend")
plt.show()
✓ [21] 193ms
```

```
country = "Italy"
df[df["Country name"] == country].plot(x="year", y="Log GDP per capita", title=f"{country} Life Ladder Trend")
plt.show()
✓ [22] 192ms
```